

© **Preview: Basic Ways of Factorization**

Factoring is converting a polynomial into a product of two or more simpler polynomials.

There are several methods for factorizing polynomials, including:

1. Factoring by GCF

2. Factoring by Special Patterns such as

2a. Difference of Two Squares Pattern:

$$a^2 - b^2 = (a + b)(a - b)$$

2b. Perfect Square Trinomial Pattern:

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

2c. Sum of Two Cubes Pattern:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

2d. Difference of Two Cubes Pattern:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

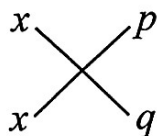
2e. Cube of a Binomial Pattern

$$a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3$$

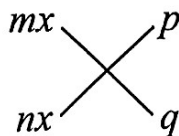
$$a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3$$

3. Factoring by Cross Method

$$x^2 + bx + c = (x + p)(x + q)$$



$$ax^2 + bx + c = (mx + p)(nx + q)$$



⊙ **Concept 1: Factoring by Grouping**

Grouping method is used to factorize polynomials that cannot be directly factored using the common factor method or formulas. Polynomials that can be factored by grouping typically have four or more terms. Generally, there are two common forms of grouping: two-two grouping and three-one grouping.

Math Exploration 1

- 1 It is known that $a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 6c + 9 = 0$. What is the product of a , b and c ?
 A. -27 B. -18 C. 0 D. 18 E. 27

- 2 Real numbers x and y satisfy the equation $x^2 + y^2 = 10x - 6y - 34$. What is $x + y$?
 (2013 AMC 10B Problems, Question #11)
 A. 1 B. 2 C. 3 D. 6 E. 8

3 Which of the following is the factor of the polynomial: $3^{n+2} - 2^{n+3} + 3^n - 2^{n+1}$?

A. 3

B. 6

C. 8

D. 10

E. 12

4 Which of the following expression is the factor of $x^2 + 8y - 4x - 4xy + 4$?

A. $x - 4$

B. $x + 2$

C. $x - 4y - 2$

D. $x - 4y$

◎ **Concept 2: Factoring by Splitting and Adding Terms (Completing Squares)**

Splitting terms: Convert a certain term into the sum of several terms.

Example: $3a^2 = a^2 + 2a^2$

Adding terms: Add several terms into the expressions without changing the value of the expressions.

Example: $a^2 + 1 = a^2 + 2a - 2a + 1$

Both splitting and adding terms will not change the value of the expression. When the expression cannot be factored directly, we can apply these methods to allow us to factor by finding common factors, or by the formulae method. If we can make connections between some terms in the original expression, we can also use the grouping method to factor.

Completing the square: We may add terms to expressions to make up important missing parts of the original polynomial to construct a perfect square in some part of the new polynomial.

After completing the square, the difference between two squares is usually used to factor the polynomial.

Method: We usually make up the middle term $2ab$, or one of the square term b^2 (or a^2). The way of completing the square depends on your observation of the problem. After that, we manipulate the polynomial to the square of difference form $A^2 - B^2$, and factor it into the form of $(A - B)(A + B)$.

Math Exploration 2

- 1 Factorize $a^3 - 4a + 3$.

2 Which of the following is the factor of the polynomial: $x^3 - 9x + 8$?

A. $x + 1$

B. $x - 1$

C. $x^2 + 1$

D. $x^3 + 1$

3 Complete the square to factorize - add and subtract terms:

(1) $x^4 + 4$

(2) $x^8 + x^4 + 1$

© **Concept 3: Simon's Favorite Factoring Trick**

Simon's Favorite Factoring Trick is a technique often used to factor expressions where completing the square or other typical methods are less intuitive. It is especially useful for factoring certain types of equations that involve two variables, particularly those that look like this:

$$xy + ax + by + ab$$

The goal is to rewrite the expression so that it can be factored as the product of two binomials.

General Formula: $xy + ax + by + ab = (x + b)(y + a)$.

Math Exploration 3

- 7 Given that x, y are positive integers and $xy - 2x - 3y = 6$, find the sum of all the possible values of x .
- 2 Given that x, y are positive integers and $6xy - 10x - 21y = 49$, find all possible solution pairs (x, y) .

- 3 Consider the set of all fractions $\frac{x}{y}$, where x and y are relatively prime positive integers.

How many of these fractions have the property that if both numerator and denominator are increased by 1, the value of the fraction is increased by 10%? (2015 AMC 10A Problems, Question #15)

- A. 0 B. 1 C. 2 D. 3
E. infinitely many

⊙ **Concept 4: Find the Extreme Value by Completing Squares**

Because of the non-negativity of perfect squares that $x^2 \geq 0$, we can find the minimum value of the polynomials by completing the squares.

Math Exploration 4

⑦ If x and y are real numbers, what is the least value of $2x^2 - 4xy + 4y^2 + 8y + 11$?

- A. 1 B. 3 C. 5 D. 7 E. 9

② What is the least possible value of

$$(x + 1)(x + 2)(x + 3)(x + 4) + 2019$$

where x is a real number? (2019 AMC 10A Problems, Question #19)

- A. 2017 B. 2018 C. 2019 D. 2020 E. 2021

1 $\frac{1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + 97^2 - 98^2 + 99^2 - 100^2}{1 + 2 + 3 + 4 + 5 + 6 + \dots + 97 + 98 + 99 + 100} = \underline{\hspace{2cm}}.$

A. -5050 B. -1 C. 0 D. 1 E. 5050

2 The expression $\frac{2021}{2020} - \frac{2020}{2021}$ is equal to the fraction $\frac{p}{q}$ in which p and q are positive integers whose greatest common divisor is 1. What is p ? (2021 AMC 10B Problems, Question #3)

A. 1 B. 9 C. 2020 D. 2021 E. 4041

3 Given that $x^2 + x + 1 = 0$. The value of $x^3 + 2x^2 + 2x + 2018$ is $\underline{\hspace{2cm}}.$

- 4 How many ordered pairs of integers (x, y) satisfy the equation $x^{2020} + y^2 = 2y$? (2020 AMC 10B Problems, Question #9)
- A. 1 B. 2 C. 3 D. 4
E. infinitely many
- 5 Given that x, y are positive integers and $xy - 4x - 5y = 14$, the sum of possible values of x is _____.
- 6 If $a + b = 6$ and $a^3 + b^3 = 72$, then the value of $a^2 + b^2$ is _____.

- 7 For how many integers x is the number $x^4 - 51x^2 + 50$ negative? (2014 AMC 10B Problems, Question #20)

A. 8 B. 10 C. 12 D. 14 E. 16

- 8 In an isosceles triangle, the two side lengths a and b satisfy the conditions:
 $9a^2 - b^2 = -13$ and $3a + b = 13$. What is the length of the third side of the triangle?

A. 2 B. 7 C. 2 or 5 D. 2 or 7 E. 5 or 7

9 Which of the following is the factor of the expression:

$$(x^4 - x^2 - 40)(x^4 - x^2 - 44) - 896?$$

A. $x + 4$

B. $x + 6$

C. x^2

D. $x - 3$

E. $x^2 + 9$

10 If $a + b + c = 6$, $a^2 + b^2 + c^2 = 14$ and $a^3 + b^3 + c^3 = 36$, $a^4 + b^4 + c^4 = \underline{\hspace{2cm}}$.